

CHTOURA, Lebanon

WMO: 401010

Lat: 33.817N

Lon: 35.850E

Elev: 3018

StdP: 13.16

Time Zone: 2.00 (LBN)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.8	33.2	15.5	13.5	52.9	20.2	16.9	51.9	18.7	41.8	16.3	45.4	1.7	50	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	27.4	98.9	65.8	96.0	65.4	93.3	64.9	70.2	89.5	69.0	88.1	67.7	86.9	7.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	102.8	76.1	62.8	95.7	77.1	60.9	89.3	77.8	36.2	89.2	35.0	88.0	33.9	86.9	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.4	12.7	26.7	102.4	3.0	3.2	24.5	104.8	22.8	106.7	21.1	108.5	19.0	110.8		
(5)			25.4	73.1	2.8	1.2	23.4	74.0	21.8	74.7	20.2	75.3	18.2	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.8	45.0	47.4	52.6	59.3	67.5	74.5	79.0	79.5	74.4	67.5	57.1	48.2
	DBStd	13.32	5.63	6.24	6.80	6.60	5.98	4.80	4.04	3.92	4.22	5.74	5.23	5.41
	HDD50	439	173	112	48	6	0	0	0	0	0	0	5	95
	HDD65	2537	621	492	386	193	43	1	0	0	0	39	242	520
	CDD50	5094	17	41	130	284	542	736	899	913	732	542	218	40
	CDD65	1716	0	0	3	21	120	287	434	448	282	116	5	0
	CDH74	20066	0	1	32	258	1408	3457	5468	5479	2840	1068	55	0
	CDH80	9677	0	0	2	36	488	1600	2946	3010	1255	337	3	0
(14) Wind	WSAvg	4.4	3.8	4.6	4.9	5.2	5.0	5.4	5.3	4.6	4.4	3.5	2.9	3.3
(15) Precipitation	PrecAvg	24.8	5.5	4.3	3.7	1.7	0.4	0.0	0.0	0.0	0.0	1.3	2.7	5.2
	PrecMax	46.2	15.9	9.6	8.9	7.7	1.6	0.2	0.0	0.0	0.6	3.6	8.7	14.7
	PrecMin	13.9	1.7	1.1	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5
	PrecStd	6.8	2.8	2.2	2.1	1.6	0.5	0.0	0.0	0.0	0.1	1.0	2.0	3.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	71.7	77.8	84.3	93.5	99.0	103.7	102.6	96.8	91.0	79.6	71.0
		MCWB	49.3	52.8	54.7	57.5	62.0	63.6	66.1	67.4	64.1	60.5	56.7	50.8
	2%	DB	60.9	66.1	72.9	80.3	89.2	94.7	99.1	99.0	93.1	87.3	75.1	64.6
		MCWB	48.0	49.8	52.7	56.5	60.6	63.3	65.5	66.8	63.2	60.0	54.3	50.1
	5%	DB	57.3	61.9	69.0	77.2	85.5	91.5	95.6	96.1	90.4	83.8	71.3	60.2
		MCWB	46.3	48.2	51.2	55.9	59.7	63.3	65.5	66.4	63.1	59.3	53.6	48.3
	10%	DB	53.8	57.9	65.4	73.7	82.0	88.7	92.7	93.2	87.3	80.0	67.4	56.7
		MCWB	45.3	46.6	49.7	54.5	58.8	62.8	65.6	66.2	62.8	58.8	52.5	47.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	53.7	57.3	60.3	64.2	68.8	72.3	72.6	69.0	64.6	59.3	54.5
		MCDB	61.4	69.0	73.2	76.3	85.7	89.4	92.4	92.5	85.3	80.9	73.5	62.3
	2%	WB	49.2	51.1	54.1	58.4	62.6	66.7	70.0	70.7	67.1	62.5	57.1	52.0
		MCDB	58.3	63.3	68.6	76.0	83.2	88.0	90.4	89.9	84.3	80.1	69.1	60.3
	5%	WB	47.6	49.3	52.5	56.8	61.1	65.3	68.6	69.5	65.6	61.2	55.4	50.3
		MCDB	55.1	59.6	66.1	74.3	81.4	86.2	88.7	88.1	83.4	77.5	66.6	56.9
	10%	WB	46.1	47.4	50.8	55.3	59.9	63.9	67.3	68.2	64.4	60.1	53.9	48.7
		MCDB	52.7	56.0	63.1	71.8	79.1	84.5	87.3	87.1	82.4	75.4	64.0	54.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	15.6	18.3	21.1	24.4	26.1	26.4	27.4	25.9	23.6	20.0	15.7
		MCDBR	21.7	23.3	24.8	26.6	28.8	29.5	30.6	30.8	29.4	28.4	24.7	22.6
	5% WB	MCWBR	11.2	10.7	10.7	11.6	11.6	11.8	12.4	11.6	11.1	10.6	10.3	11.3
		MCWBR	18.4	20.4	22.4	24.7	26.5	27.0	27.6	28.7	26.4	24.7	20.8	17.6
(40) Clear-Sky Solar Irradiance	taub		0.356	0.393	0.445	0.500	0.492	0.414	0.422	0.433	0.418	0.435	0.381	0.362
		taud	2.255	2.128	1.974	1.849	1.908	2.168	2.151	2.116	2.146	2.071	2.209	2.240
	Ebn at Noon		266	267	261	253	256	276	273	268	267	248	253	254
		Edh at Noon	34	43	55	65	62	48	48	49	45	45	36	33
(44) All-Sky Solar Radiation	RadAvg		820	1085	1554	1949	2336	2604	2548	2310	1946	1425	1020	773
	RadStd		71	109	96	119	112	43	34	47	55	72	66	69

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.54	N/A	-2.39	N/A	N/A	N/A	N/A	N/A	+311	+198

Nomenclature: See separate page